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Artificial Intelligence–based algorithms on social media:
The spread of misinformation and disinformation and lack of new content discoverability

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Introduction

Since the birth of the Internet in the 1980s, there has been a shift in the way people consume media content in a digitized landscape that disrupts the traditional media industry. People now have easy access to a vast amount of information online, especially through devices such as smartphones and online media platforms like social media.

This exposes users to an overwhelming amount of information in the online environment which may become difficult to sort through to find relevancy in the material shared online. Previous studies suggests that overexposure to this information overload online would lead to a reduction the ability of users to comprehend the entire context of the information, their media literacy, and critical thinking skills (Pelau, Pop, Ene, & Lazar, 2021).

Therefore, to make the process of finding suitable content for oneself online, the integration of Artificial Intelligence (AI) tools that study users' individual behaviors, preferences, and interests with social media helps in curating a personalized online experience.

According to Russell and Norvig (2010), artificial intelligence can be defined as computer systems that mimic the human thought process involved in problem solving, learning, or speech. In other words, AI is the science that deals with trying to imbue human intelligence into computer systems to perform tasks that require cognitive processes like reasoning, perception, and application of knowledge which could only be done by humans before.

Companies benefit from using AI with social media to study more about their customers by gathering and analyzing their activity on social media and make inferences about their social behaviors and buying patterns through big-data analytics (Sadiku et al., 2021). A common use of AI in social media is the use of facial recognition on platforms like Snapchat, Instagram, and TikTok to make face filters work; but a more important role can be seen in its ability to curate recommender algorithms on social media where techniques like machine learning and natural language processing analyzes patterns and trends in users' behaviors to learn their preferences (Frąckiewicz, 2023).

AI's effectiveness in curating a personalized online social media experience for users to deal with the issues of decision making in the face of information overload cannot be ignored due to its ability to increase user satisfaction and overall online experience. However, it is also important to remember that despite the advantages of AI recommender systems in social media, research has also shown that it comes with obstacles such as biased algorithms which eventually lead to the creation of filter bubbles and media echo chambers which can lead to the spread of misinformation and disinformation (Trattner et al., 2021).

This paper aims to connect the implementation of artificial intelligence in recommender-based algorithms to increase in selective exposure of content where cognitive dissonance and confirmation bias play a role in the spread of misinformation and disinformation. This selective exposure would also lead to a reduction in the discoverability of new content, thereby, making AI recommender algorithms part of the reason for a lack of new content discoverability.

Background

AI recommender systems

To understand the relation between how AI recommender algorithms can lead to selective exposure, it is important to understand the definitions of both terminologies. Recommender systems or algorithms are defined as a part of machine learning, aiming to predict what consumers are looking for based on studying users' behavioral data and intent and eventually lead them to other media content that is similar to the content they have engaged with in the past (Dooley, 2023).

McKelvey and Hunt (2019) say that various media platforms differ in the way users can discover content. They provide a three-concept framework of surrounds (the way choices are made available on the platform), vectors (the ways software guides a user through the choices), and experience (the overall online experience of discovering and consuming media content). Using this framework, it can be said that AI recommender algorithms on social media are a vector that leads to discoverability of content a person is more likely to engage with online, thereby increasing the value of their online experiences.

It is important to remember that only a few platforms, like TikTok, employ a sole recommendation algorithmic model but, recently, there is a new shift to use such algorithms in other social media platforms. Especially after the success of TikTok's *For Your Page*, where videos that relate with the content a user engages with, appear one after the other leading to higher user engagement on the app.

Other platforms started making use of the recommender systems like Instagram's *Explore Feed*, X's (formerly known as Twitter) suggestions of tweets one might like, and Facebook's *News Feed* algorithm (Narayanan, A. (2023). Therefore, even if there are slight differences in the way each app shows content, all social media platforms are starting to exhibit heavy reliance on recommender algorithms making the user experience across all platforms similar.

Selective exposure

Sears and Freedman's (1967) critical analysis of selective exposure is a classic research paper that suggesting the best fitting definition of selective exposure is the idea that people usually like to expose themselves to media content that will align with already existing notions of their mind.

It explores the concept that people may actively or unconsciously seek out content that is in line with their set of beliefs to reinforce their own points of view and disregard content that opposes their views. This can be explained as cognitive dissonance which Festinger (1957) defines as the psychological discomfort one feels when they encounter information that goes against their set of beliefs. Thus, the idea of selective exposure came about as a solution to avoid this discrepancy of cognition. Subsequent research, such as Bennett and Iyengar (2008), has since established a strong link between selective exposure and political communication which may in turn lead to confirmation and ingroup biases.

Knobloch-Westerwick et al. (2017) says that users avoid information that does not fit with their logic (cognitive dissonance) and instead favor content that is consistent with their already-present attitudes in their mind and reconfirms their choices (confirmation bias) which then leads to polarization of groups such as ingroups and outgroups online.

The creation of recommender algorithms on social media through machine learning means the addition of an entire new layer to the theory of selective exposure. Selective exposure does not necessarily mean that users would never engage with content inconsistent with their beliefs, but just that it was more unlikely to happen (Seddik et al., 2023).

People consuming content that only reaffirmed their beliefs would lead to them isolating themselves in filter bubbles and echo chambers and these algorithms that select content for the users to see only heightens the effect of selective exposure as users get stuck in a self-reinforcing loop of information that they can agree with.

Where people once either consciously or unconsciously made the decision to interact with certain media content, the advent of these AI algorithms means that the decision-making process for choosing which content to interact with is made by the computer based on the user's preferences. But research such as Gunn (2021) differs, saying that although there are concerns surrounding the generation of filter bubbles due to algorithmic selection of media presenting an individualized presentation of choices, evidence from empirical research usually concludes the opposite is true, i.e., such algorithms lead to diversification of media content. However, in this caset the research is mainly focused on search engines rather than social media.

Selective exposure through AI and misinformation and disinformation

Studies has shown that a link can be established between the concept of selective exposure and the spread of misinformation and disinformation, but first it is necessary to distinguish between the two terms. Misinformation can be simply defined as the presentation of distortion of information, but usually as a mistake. Whereas disinformation is where false information has been deliberately used in order to send out a certain message to the audiences (Scheufele and Krause, 2019).

An example where selective exposure could lead users into a cycle of engaging with misinformation is presented in Guess and Reifler's (2018) paper leading up to the 2016 U.S. Presidential Elections. They found that those who supported Donald Trump as a candidate were likelier to visit fake news websites that were also pro-Trump and the same applied for pro-Clinton supporters. Although pro-Trump supporters consumed more fake news articles than pro-Clinton supporters, the results of the study supported the paper's hypothesis that, depending on their candidate preferences, both parties visited fake-news websites that supported their respective candidates.

Along with the benefit of developing connections all around the world through social media, there are also concerns with how such platforms become a breeding ground of misinformation and disinformation, thereby making it imperative to study what leads to the spread of these false narratives online.

Benzie and Montasari (2022) studied how AI machine learning in social media leads to the distribution of inaccurate and harmful information online. They examined the use of AI software robots with programmed algorithms assists in spreading information by directly leading certain content to reach the feeds of social media users. Their study supports the view that such algorithms bring like-minded information a person would agree with on their online feeds and create bubbles of information or echo chambers.

The information that is spread depending on user preferences are not checked for facts by bots as their programming is concerned with just delivering content that the person will likely engage with. And when more than half users spend less than 15 seconds on a page, they are not likely to recheck the information that is directly placed on their feeds. Since the content is going to be aligned with their belief system due to AI algorithms, they would not question the integrity of the content as it would imply them having to question their own belief system which give psychological discomfort.

Although AI bots involved with algorithm planning here, are not specifically designed to help the spread of false and malicious information,rather they are designed to help users navigate their way to media that aligns with their ideas, even if it may be intentionally or unintentionally misleading.

Discussion

The Background set up the thesis that AI recommender-based algorithms can eventually lead to the spread of misinformation and disinformation online due to cognitive dissonance and selective exposure. Is this notion reinforced by recent world events?

In this section, the spread and consumption of misinformation and disinformation can be observed in two examples, i.e., the "infodemic" during the COVID-19 pandemic and the difficulty in distinguishing fact from fiction on social media in regard to the current Israel/Hamas conflict in Gaza.

COVID-19

Amongst growing health concerns during the COVID-19 global pandemic, a multitude of other issues such as the sale of bogus drugs, fake news, and misinformation about treatment surfaced.

While there were reputable organizations putting out credible information about the novel coronavirus to counter the spread of misleading information online, new information was being released to the public at a slower rate (Amin et al., 2021). The heightened tensions of public over the scarcity of information on the virus during the early stages of the pandemic made people take it upon themselves to share information they felt was right in the treatment and prevention of the virus. For example, talk of heat killing COVID-19 and false updates on the development of the vaccine dominated social media spaces.

This led to a large amount of fake news going viral to the point where even the World Health Organization pointed out the simultaneous infodemic that was occurring. In Amin et al. (2021) distributed a questionnaire to 112 participants to determine how attention-based design online would lead to spread of disinformation. The paper found that how people care about consuming information that stays true to their ideologies was due to the attention-based design of social media platforms. This played a significant part in influencing people to engage and further share inaccurate content.

Another study by Fernández and Cantador (2021) studied datasets on misinformation posts on X (formerly known as Twitter). They concluded that research into algorithmic models on social media and the connection to the spread of misinformation is multi-faceted and require further research.

Israel/Hamas conflict

The second example follows posts that have recently circulated on social media all around the world after Hamas's attacks on Israel on October 7th, 2023. In retaliation, the Israeli government declared "a war against Hamas" with military strikes bombarding Palestinian occupied territory in the Gaza Strip. Amidst the conflict social media spaces have seen a clear-cut polarization of two major groups: one in support of the Israeli government and the other in favor of the Palestinian liberation movement. There are examples of disinformation arising from both groups with AI playing a role in not only the consumption of such content in terms of algorithms, but also in the generation of them.

One example of content circulating on pro-Israeli spaces on social media, is the image of a charred baby shared by American conservative commentator, Ben Shapiro on X. However, the tweet is now claimed to be generated by AI and was slammed as the intentional spread of disinformation to gather public support for Israel's actions in Gaza. Another video of Hamas paragliders jumping on a sports field was shared on social media where people were told that this was taken on the day of the October 7 attacks and had been reposted more than 2,800 times on X and had over 38,000 views on TikTok. This video was later debunked associated with a false narrative by APNews as it was a video from September taken of parachute jumpers in Cairo, Egypt.

Content that supports Israel is shown on the feeds of pro-Israeli supporters while content that supports Palestine shows on the feeds of pro-Palestine supporters, thus, people on both sides are mainly engaging with content that caters to their beliefs even in cases of blatant misinformation and disinformation being spread. AI algorithms create filter bubbles around the two groups so much so that they believe all the information the in-group has is true and that provided by the outgroup is fake and incorrect.

The use of AI in the digital landscape is still a fairly recent phenomena but one that is evolving at a rapid pace. This paper draws upon evidence presented in previous literature and attempts to show practical application of how recommendation algorithms based on AI machine learning leads to the spread of misinformation and disinformation and eventually reduces the discoverability of new content for social media users.

However, there are conflicting views over the fear of AI algorithms leading to echo chambers and filter bubbles. Which is why more in-depth research studying direct links between recommender system algorithms and reduction in new content availability would be helpful. Most research at present concentrates on the benefits of artificial intelligence when it comes to content discoverability but only touches upon the advantages of having user-preferences based algorithms.

While it is necessary to acknowledge the ease such AI systems have provided for online users, it is equally important to highlight the flaws of implementing modern technology with regards to other concepts like attention-based design and its association with decreasing media literacy, increased screen time, etc. Only by doing extensive research into the flaws of AI algorithms can proposals for solutions be developed to resolve such issues.

Implications

The discussion of artificial intelligence's role in creating recommendations algorithms for social media that uses collaborative filtering of media content to present netizens with a personalized curation of their ideal online experience raises questions on whether this machine-done filtering of media is good for online users.

People no longer have time to wade through the insurmountable pile of information that exists on social media, and while AI algorithms make it easier for users to enjoy content, because the decision-making is being done by an external source, this creates the illusion of free choice. The result of this is the omission of media content the user is not likely to interact with. This, in turn, leads to the reduction of new content being made available.

Therefore, the lack of exposure to new and conflicting content for social media users, restricts their view of any issue and leads to uninformed and biased decison-making.

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